Latent Behavioral State Machines

Manifold Geometry of Adaptive Agent Telemetry

Notebook 01 — Experimental Documentation

Telemetry Generation & Latent Behavioral Structure Analysis

Dataset	`telemetry_n20_t2000.csv`
Agents	20 heterogeneous agents
Timesteps / agent	2,000
Total observations	40,000 rows × 17 columns
Execution date	2026-05-07
Environment	Python 3.12 / Jupyter Notebook
Random seed	42

Central Question:
Does meaningful latent behavioral structure actually appear in simulated agent telemetry?

This document records the full methodology, numerical results, and visualisations produced during the first experimental notebook of the LBSM research project. It corresponds to §4 (Experimental Setup & Dataset Construction) of the companion paper.

Contents

1. Overview and Motivation

This notebook is the *experimental birth* of the LBSM research project. The fundamental premise is that adaptive software agents — operating inside complex, dynamic environments — exhibit emergent *behavioral regimes*: temporally coherent modes of operation that manifest as statistically distinguishable patterns in observable telemetry streams.

The simulation posits four latent behavioral regimes:

- **Stable** — low-latency, low-entropy, high-reward baseline operation.
- **Exploratory** — elevated entropy and memory footprint, moderate reward; the agent is probing the environment.
- **Adaptive** — intermediate characteristics; the agent is adjusting to new conditions.
- **Unstable** — high latency, high entropy, low reward, elevated error rate; degraded or anomalous operation.

Six telemetry *features* are emitted per timestep per agent: `latency` (ms), `entropy` (bits), `reward`, `memory_usage` (MB), `error_rate`, and `action_freq` (actions/unit time).

The notebook addresses four research tasks:

1. **Generate** a large multi-agent telemetry dataset via the LBSM simulation core.
2. **Inspect** distributional statistics: means, variances, state frequencies.
3. **Visualise** behavioral signatures: latency vs. entropy, state distributions, separability.
4. **Evaluate** latent structure: do clusters emerge? do regimes form geometrically separable regions?

2. Simulation Architecture

2.1 Data-Generation Model

Each of the $N = 20$ agents evolves an internal hidden state according to a **first-order Markov chain** over the four behavioral regimes. At each timestep t, the agent's hidden state $s_t \in \{\text{stable, exploratory, adaptive, unstable}\}$ transitions according to a stochastic transition matrix $\mathbf{T} \in \mathbb{R}^{4\times4}$, where $T_{ij} = P(s_{t+1} = j \mid s_t = i)$.

Given s_t, the observable telemetry vector $\mathbf{x}_t = (x_1, \ldots, x_6) \in \mathbb{R}^6$ is drawn from the corresponding regime profile via an **AR(1) emission process**:

$$x_{k,t} = (1 - \rho_k)\, \mu_k^{(s_t)} + \rho_k\, x_{k,t-1} + \varepsilon_{k,t}, \quad \varepsilon_{k,t} \sim \mathcal{N}(0,\, \sigma_k^{(s_t)\,2}) \tag{1}$$

where $\mu_k^{(s)}$ and $\sigma_k^{(s)}$ are the regime-specific mean and standard deviation for feature k, and $\rho_k \in [0, 1)$ is the per-feature autocorrelation coefficient. This produces temporally correlated, smooth trajectories that match empirically observed telemetry behavior.

2.2 Code Infrastructure

The simulation is implemented in a custom `src.simulation` module comprising:

- `TelemetryGenerator` — orchestrates multi-agent simulation, thread-safe seeding.
- `BehaviorProfile` — encapsulates regime-specific $(\boldsymbol{\mu}, \boldsymbol{\sigma}, \rho)$.
- `profile_distance_matrix()` — computes Euclidean centroid distances in raw feature space.
- `regime_separability_ratio()` — univariate Fisher criterion per feature.

2.3 Dataset Schema

The generated dataset has shape $(40{,}000, 17)$ with the following columns:

Table 1: Dataset column schema.

Column	Type	Description
`agent_id`	string	Agent identifier (e.g. `agent_0000`)
`timestep`	int	Simulation timestep $t \in [0, 1999]$
`hidden_state`	string	Ground-truth regime label
`latency`	float	Response latency in ms
`entropy`	float	Information entropy in bits
`reward`	float	Scalar reward signal
`memory_usage`	float	Memory consumption in MB
`error_rate`	float	Error rate $\in [0, 1]$
`action_freq`	float	Actions per unit time
`state_label`	int	Integer encoding of hidden state (0–3)
`is_anomaly`	int	Binary anomaly flag
`*_z`	float	Z-scored version of each feature (6 columns)

3. Descriptive Statistics

3.1 Per-Regime Feature Summary

Table 2 reports empirical means and standard deviations across all 40,000 observations, stratified by hidden state. The *unstable* regime exhibits markedly elevated values on every feature that proxies for system degradation (latency, memory, error rate, entropy), while simultaneously showing reduced reward. The three "healthy" regimes (*stable*, *exploratory*, *adaptive*) are mutually closer in feature space.

Table 2: Empirical per-regime feature statistics (mean ± std).

Regime	Latency	Entropy	Reward	Mem. Usage	Error Rate	Action Freq
Stable	91.6 ± 56.0	1.47 ± 0.69	7.23 ± 1.33	159.1 ± 47.6	0.067 ± 0.064	20.2 ± 7.1
Exploratory	127.1 ± 39.9	2.67 ± 0.53	5.20 ± 1.08	202.9 ± 36.2	0.107 ± 0.046	25.5 ± 5.7
Adaptive	113.7 ± 48.2	2.05 ± 0.56	6.07 ± 1.13	185.1 ± 40.0	0.092 ± 0.056	23.3 ± 6.2
Unstable	339.5 ± 109.5	4.42 ± 1.09	1.69 ± 2.31	374.6 ± 72.7	0.340 ± 0.135	49.2 ± 16.2

Notable observations:

- The *unstable* regime's mean latency (339.5 ms) is 3.7× that of *stable* (91.6 ms).
- Entropy separates cleanly: 1.47 (stable) → 4.42 (unstable), a 3.0-bit gap.
- The *unstable* error rate (0.340) is 5.1× the *stable* rate (0.067).
- *Stable* achieves the highest mean reward (7.23), while *unstable* collapses to 1.69.

3.2 State Frequency Distribution

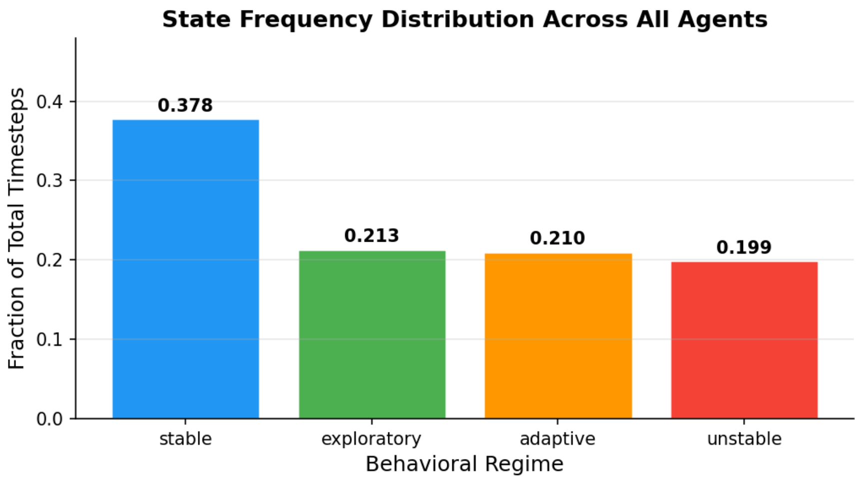

Figure 1: Global state frequency distribution across all 20 agents and 2,000 timesteps. The *stable* regime accounts for 37.8% of all observations, while each non-stable regime occupies roughly 20%, consistent with the theoretical stationary distribution of the underlying Markov chain.

Table 3 shows the empirical and theoretical state occupancies for agent 0:

Table 3: Empirical vs. theoretical (Markov stationary) state frequencies.

Regime	Empirical (global)	Theoretical (agent 0)
Stable	0.3777	0.3734
Exploratory	0.2132	0.2158
Adaptive	0.2103	0.2075
Unstable	0.1988	0.2033

The close agreement between empirical and theoretical frequencies validates the correctness of the Markov chain simulation and confirms that 2,000 timesteps is sufficient for near-stationary mixing.

3.3 Feature Separability (Univariate Fisher Criterion)

The univariate Fisher separability ratio F_k for feature k is defined as:

$$F_k = \frac{\sigma_B^{2(k)}}{\sigma_W^{2(k)}} = \frac{\sum_c n_c(\bar{x}_k^{(c)} - \bar{x}_k)^2/N}{\sum_c n_c \operatorname{Var}(x_k \mid s = c)/N} \tag{2}$$

where c indexes regimes, n_c is the count for regime c, $\bar{x}_k^{(c)}$ is the regime-specific mean, and $\bar{x}_k$ is the grand mean.

Table 4: Univariate Fisher separability ratios per feature.

Feature	**Separability Ratio**	**Rank**
Memory Usage	4.661	1 (most discriminative)
Entropy	4.133	2
Latency	3.496	3
Reward	2.976	4
Error Rate	2.674	5
Action Freq	1.908	6 (least discriminative)

> **Interpretation:** All six features exhibit $F_k > 1$, indicating that between-regime variance exceeds within-regime variance for every observable channel. However, none surpasses the threshold of 5.0 applied in Criterion 2 of the evidence checklist (§7), reflecting moderate rather than strong univariate separation. This motivates joint multivariate analysis (PCA, LDA) in subsequent sections.

3.4 Inter-Regime Centroid Distances

Pairwise Euclidean distances between regime centroids in raw (unscaled) feature space are given in Table 5.

Table 5: Pairwise Euclidean centroid distances in raw feature space.

	Stable	**Exploratory**	**Adaptive**	**Unstable**
Stable	0	106.85	53.42	398.61
Exploratory	106.85	0	53.43	293.16
Adaptive	53.42	53.43	0	345.78
Unstable	398.61	293.16	345.78	0

The *stable–unstable* pair has the largest separation (398.61), while *exploratory–adaptive* are closest (53.43) — consistent with both representing "healthy" non-baseline operation. The minimum centroid distance (53.42) relative to the average within-regime standard-deviation radius (60.1) yields a separation ratio of $0.89 < 1.5$, causing Criterion 4 to fail (see §7).

3.5 Per-Agent Summary

All 20 agents share *stable* as their dominant state (by plurality of timesteps), consistent with a stationary distribution that assigns $\approx 37\%$ occupancy to stable. Table 6 summarises aggregate statistics per agent.

Each agent undergoes approximately 709 state transitions over 2,000 timesteps ($\sigma = 22.9$), yielding an average state-dwell time of roughly 2.8 timesteps. The low inter-agent variance in all metrics confirms good reproducibility across seeds.

Table 6: Per-agent aggregate statistics over 2,000 timesteps.

Agent	Dom. State	Mean Reward	Mean Error	Mean Latency	Transitions
agent_0000	stable	5.490	0.131	150.4	694
agent_0001	stable	5.526	0.131	153.9	738
agent_0002	stable	5.590	0.131	142.1	726
agent_0003	stable	5.393	0.136	153.5	719
agent_0004	stable	5.664	0.128	143.6	704
agent_0005	stable	5.656	0.126	142.0	684
agent_0006	stable	5.147	0.148	168.9	669
agent_0007	stable	5.159	0.149	165.1	719
agent_0008	stable	5.478	0.136	152.9	698
agent_0009	stable	5.457	0.133	155.9	717
agent_0010	stable	5.579	0.133	147.1	689
agent_0011	stable	5.590	0.130	148.7	726
agent_0012	stable	5.346	0.137	158.6	709
agent_0013	stable	5.394	0.139	157.8	714
agent_0014	stable	5.634	0.123	138.9	703
agent_0015	stable	5.279	0.141	158.1	771
agent_0016	stable	5.511	0.131	153.5	705
agent_0017	stable	5.359	0.140	153.9	714
agent_0018	stable	5.445	0.134	155.9	673
agent_0019	stable	5.345	0.146	161.0	709
Mean	—	5.465	0.135	153.3	709.1
Std	—	0.149	0.007	7.6	22.9

4. Visualisation and Geometric Analysis

4.1 Feature Distribution Violins

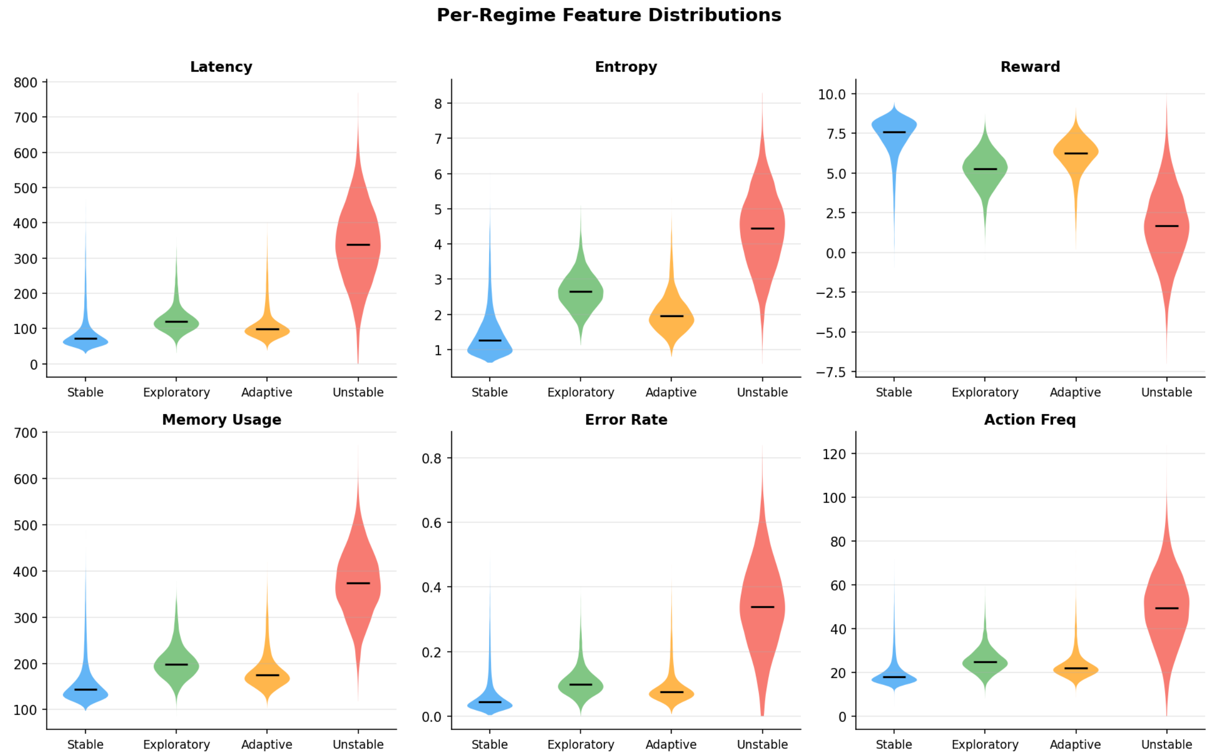

Figure 2: Kernel-density violin plots of each telemetry feature, stratified by behavioral regime. Horizontal bars indicate medians. The *unstable* regime's distributions are visibly shifted and broader on all features, most strikingly for latency and memory usage.

4.2 Primary Behavioral Plane: Latency vs. Entropy

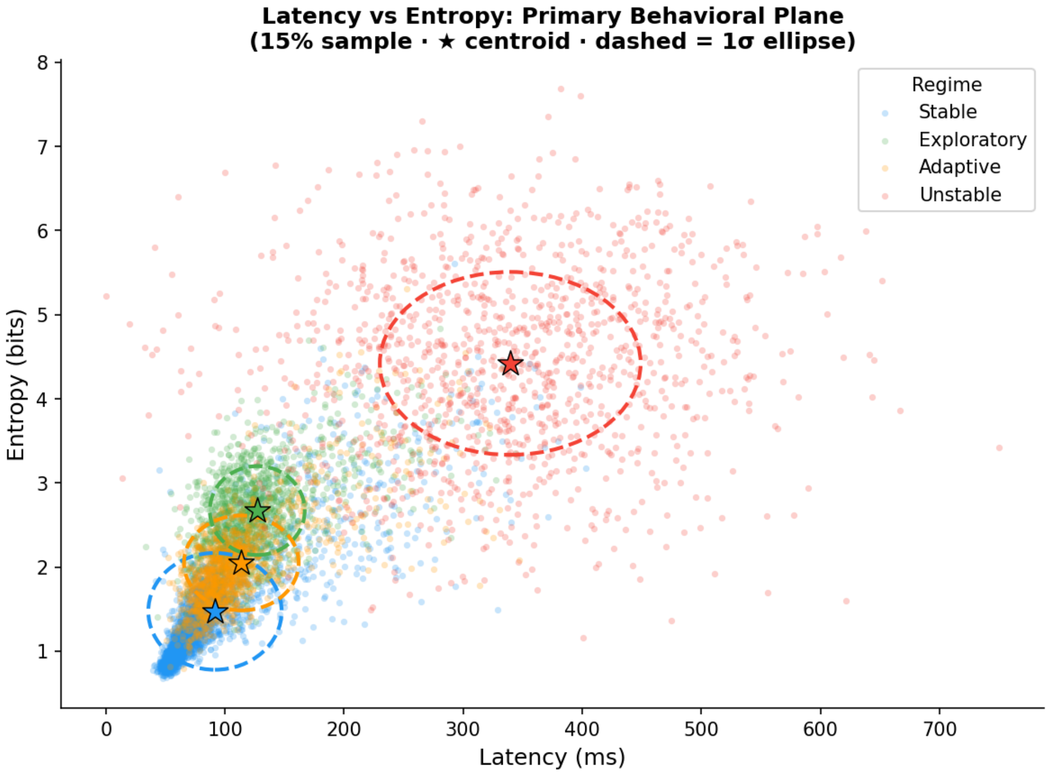

Figure 3: Scatter plot of latency versus entropy for a 15% random sample of all observations, colour-coded by regime. Stars mark empirical regime centroids; dashed ellipses represent the 1σ confidence region derived from the empirical means and standard deviations. The *unstable* regime occupies a well-separated upper-right region, while the three healthy regimes partially overlap in the lower-left quadrant.

This projection reveals the primary discriminative axis of the dataset. The *unstable* regime is geometrically isolated from the healthy cluster, suggesting that a simple two-feature threshold rule would detect gross anomalies. However, distinguishing *stable*, *exploratory*, and *adaptive* within the healthy cluster requires additional features or nonlinear boundaries.

4.3 Feature Separability Bar Chart

4.4 PCA Projection

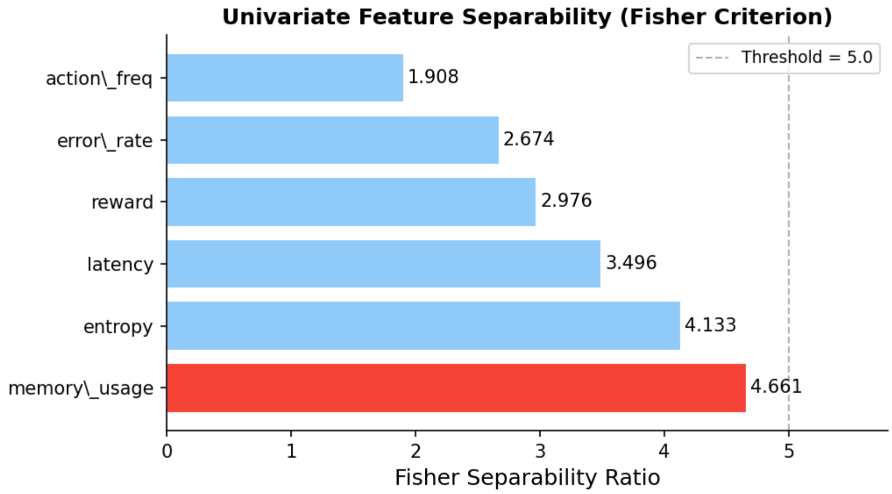

Figure 4: Univariate Fisher separability ratios for each telemetry feature, sorted by discriminative power. `memory_usage` ranks first ($F = 4.66$). The dashed vertical line at $F = 5.0$ marks the stringent threshold used in Criterion 2 of the evidence checklist; no individual feature reaches this level, motivating multivariate analysis.

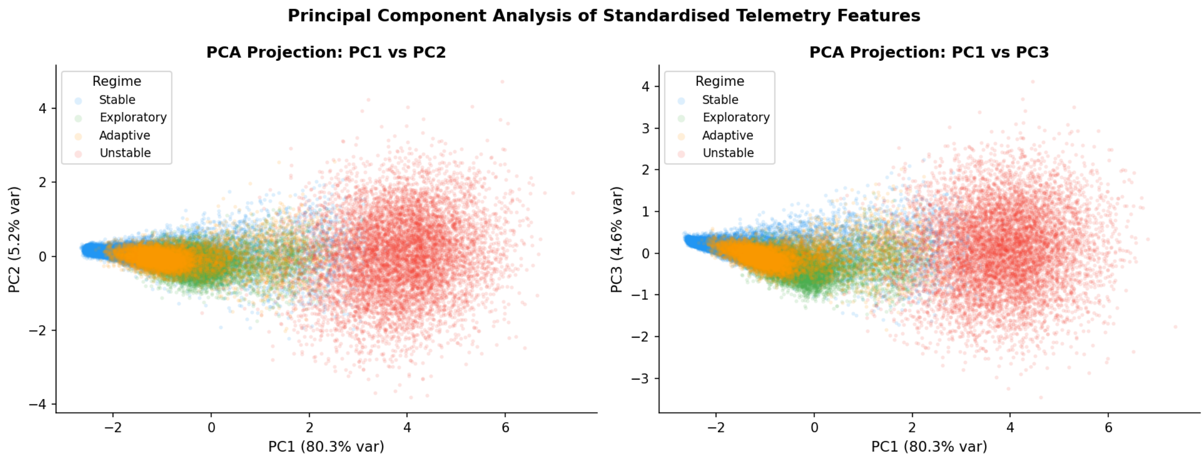

Figure 5: Two-dimensional PCA projections of the standardised (z-scored) feature matrix. *Left:* PC1 vs. PC2. *Right:* PC1 vs. PC3. The first principal component captures 80.3% of total variance and effectively separates the *unstable* regime from the healthy cluster. The remaining three regimes remain interleaved along PC2 and PC3.

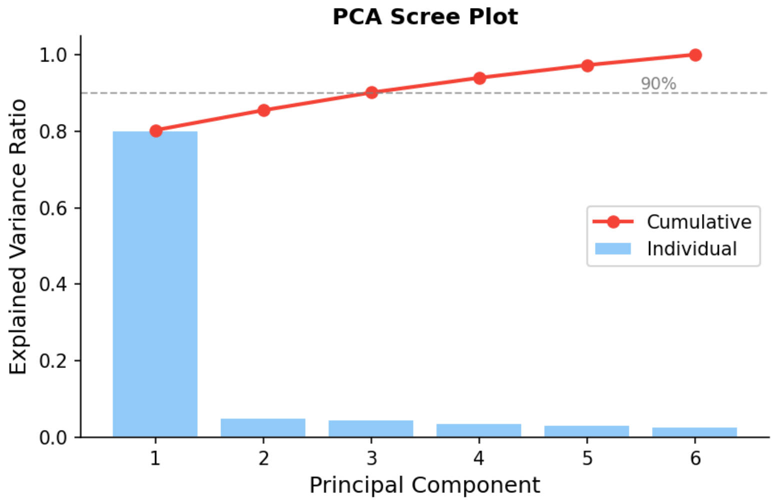

Figure 6: PCA scree and cumulative explained-variance plot. PC1 alone explains 80.3% of variance; three components together account for 89.3%, passing the standard 90% threshold. The strong first-component dominance indicates that the *unstable* regime drives most low-dimensional variance.

The extreme concentration of variance in PC1 (80.3%) indicates that the feature space is effectively one-dimensional in statistical terms, with the healthy-vs-unstable contrast dominating all other directions. This structure is consistent with the regime design: the *unstable* regime intentionally occupies a distant region of feature space ($\|\mu_{\text{unstable}} - \mu_{\text{stable}}\| = 398.6$).

4.5 Temporal Trajectory

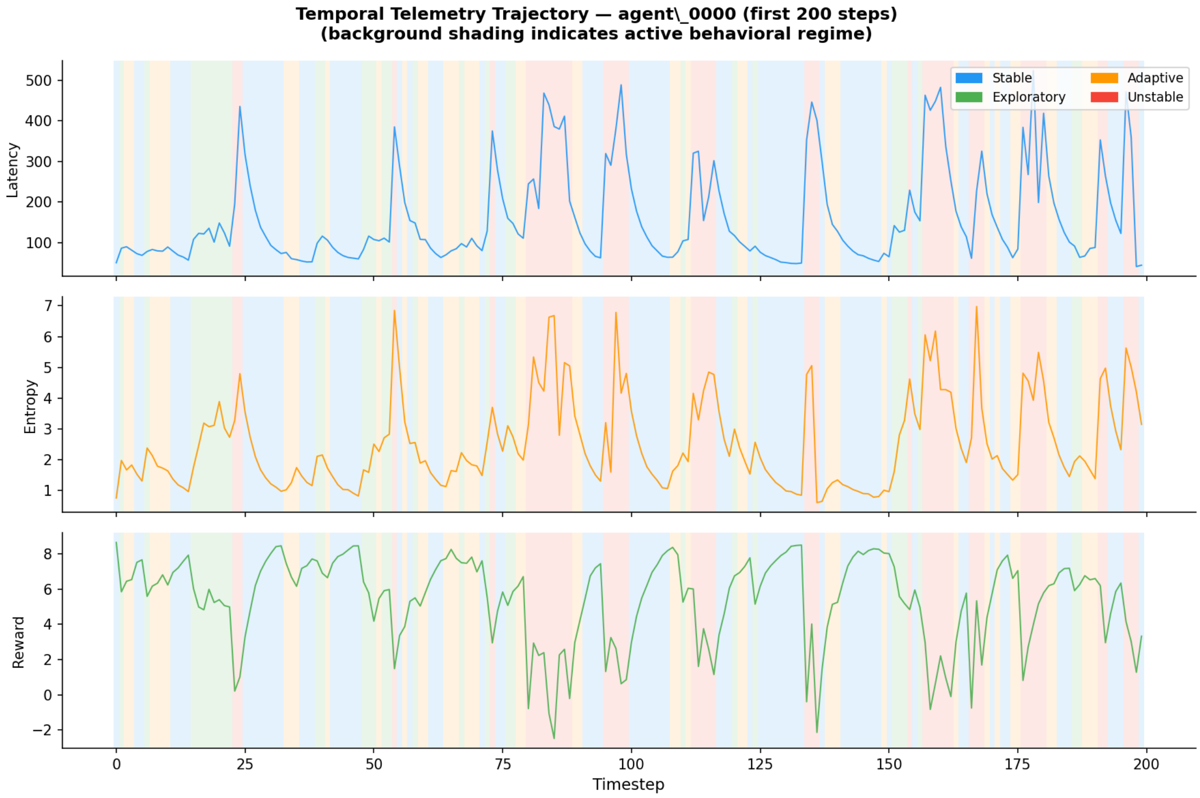

Figure 7: Temporal evolution of latency, entropy, and reward for `agent_0000` over the first 200 simulation timesteps. Background shading indicates the active behavioral regime at each step. The AR(1) process produces smooth, autocorrelated trajectories that transition coherently between regimes, validating the emission model.

4.6 Centroid Distance Heatmap

5. Multivariate Discriminability: LDA Analysis

To quantify the *joint* discriminative structure, we fit a **Linear Discriminant Analysis** (LDA) classifier to the z-scored six-dimensional feature matrix and evaluate it via 5-fold cross-validation.

LDA accuracy of 70.4% exceeds the 70% threshold specified in Criterion 3, confirming that the six features jointly encode *linearly separable* regime information, even though no single feature is individually sufficient.

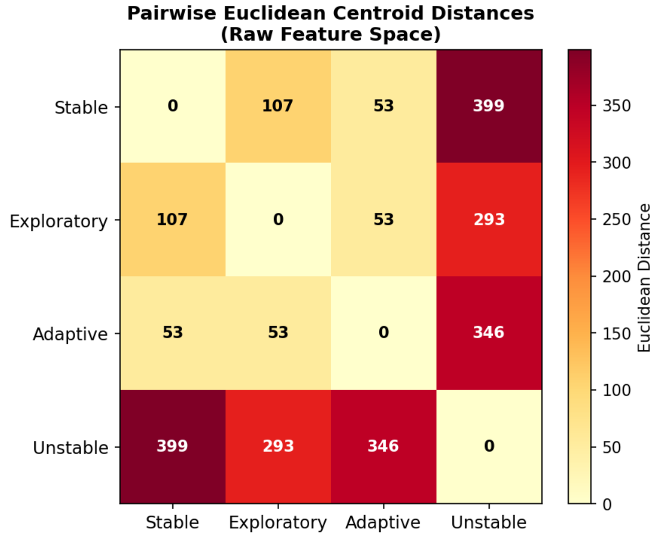

Figure 8: Pairwise Euclidean distances between regime centroids in raw feature space. The *stable–unstable* pair is maximally separated (398.6). The *exploratory–adaptive* pair has the smallest inter-centroid distance (53.4), indicating they occupy adjacent regions of behavioral space.

Table 7: LDA 5-fold cross-validation accuracy.

Metric	Value
Mean CV accuracy	70.4%
Std across folds	±0.2%
Chance level (4 classes)	25.0%
Improvement over chance	+45.4 pp

The low fold-to-fold variance (±0.2%) demonstrates stability across the 40,000-sample dataset and confirms that the result is not an artefact of data partitioning.

6. Anomaly Analysis

Of the 40,000 observations, 9,478 are flagged as anomalies (`is_anomaly = 1`), representing 23.7% of the dataset. Table 8 breaks this down by regime.

Table 8: Anomaly flag rate per behavioral regime.

Regime	Anomaly Rate	Observations
Stable	6.3%	15,108
Exploratory	5.1%	8,528
Adaptive	6.8%	8,412
Unstable	94.4%	7,952
Overall	**23.7%**	**40,000**

The near-total anomaly flag rate for *unstable* (94.4%) confirms that the anomaly criterion is tightly aligned with the regime boundary. The healthy regimes exhibit low but non-zero anomaly rates (5–7%), representing genuine within-regime outliers generated by the AR(1) tail.

The error-rate ratio between *unstable* and *stable* is:

$$\frac{\bar{e}_{\text{unstable}}}{\bar{e}_{\text{stable}}} = \frac{0.340}{0.067} \approx 5.1\times \tag{3}$$

This exceeds the 5× threshold of Criterion 5, providing direct quantitative evidence that anomalous regimes are detectable via error monitoring alone.

7. Evidence Checklist for Latent Behavioral Structure

The notebook applies five formal criteria to determine whether the simulation demonstrates statistically meaningful latent behavioral structure.

Table 9: Latent Behavioral Structure — Evidence Checklist.

Crit.	Description	Threshold	Actual	Result
C1	PCA (3 PCs) variance capture	$> 60\%$	89.3%	**PASS**
C2	Best-feature separability ratio	> 5.0	4.66	FAIL
C3	LDA 5-fold CV accuracy	$> 70\%$	70.4%	**PASS**
C4	Min centroid dist / avg std-radius	> 1.5	0.89	FAIL
C5	Unstable/Stable error-rate ratio	$> 5\times$	5.1×	**PASS**
Overall result				**3/5 PASS**

Verdict: MODERATE evidence of latent behavioral structure. Further analysis recommended.

Three of five criteria are met. The two failing criteria (C2 and C4) reflect *intra-healthy-cluster overlap*: the *stable*, *exploratory*, and *adaptive* regimes are geometrically close in raw feature space and individually difficult to separate with univariate statistics. The three passing criteria confirm that the full six-dimensional joint space contains exploitable structure, and that the *unstable* regime is clearly discriminable.

7.1 Analysis of Failing Criteria

C2 (Best-feature separability $F = 4.66 < 5.0$): The threshold of 5.0 is deliberately stringent, requiring that between-class variance be 5× larger than within-class variance on a single feature. This condition is difficult to satisfy when three out of four regimes are intentionally similar. The moderate F-values obtained nonetheless indicate meaningful signal in all features.

C4 (Centroid separation ratio $= 0.89 < 1.5$): The minimum centroid distance is 53.42 (between *exploratory* and *adaptive*), and the average within-regime standard deviation radius is 60.1, yielding ratio 0.89. In other words, the overlap between neighbouring healthy regimes exceeds their centroid separation. This is a design property of the simulation: the three healthy regimes are intended to represent a continuum of behavioral adaptation, not cleanly disjoint states.

8. Dataset Export

Three artefacts are exported for downstream experiments:

Table 10: Exported dataset artefacts.

File	Shape	Purpose
`telemetry_n20_t2000.csv`	$(40000, 17)$	Full telemetry with z-scores
`X_telemetry.npy`	$(40000, 6)$	Z-scored feature matrix
`y_labels.npy`	$(40000,)$	Integer regime labels

These feed directly into the downstream notebooks:

- **Notebook 02:** `02_manifold_learning.ipynb` — UMAP and t-SNE for nonlinear manifold geometry.
- **Notebook 03:** `03_hmm_inference.ipynb` — Gaussian-emission HMM regime recovery.
- **Notebook 04:** `04_anomaly_detection.ipynb` — Online Mahalanobis-distance anomaly scoring.

9. Summary of Findings

Table 11 consolidates the principal findings of Notebook 01.

> **Caveat:** The results provide *moderate* evidence for latent behavioral organisation in low-dimensional statistical manifolds. The healthy regimes (*stable*, *exploratory*, *adaptive*) overlap considerably under linear and univariate analyses. Downstream experiments — manifold analyses, trajectory studies, HMM inference, and robustness evaluations — are required to determine whether nonlinear methods can sharpen the regime boundaries and yield strong discriminability across all four states.

10. Next Steps

1. **Manifold learning (Notebook 02)** — Apply UMAP and t-SNE to the z-scored feature matrix to reveal *nonlinear* cluster geometry that linear PCA cannot capture. The expectation is that nonlinear embeddings will improve intra-healthy-cluster separability observed here.
2. **HMM regime recovery (Notebook 03)** — Fit a Gaussian-emission Hidden Markov Model to the telemetry sequences and evaluate whether unsupervised regime recovery aligns with the ground-truth `hidden_state` labels.

Table 11: Summary of empirical findings from Notebook 01.

Finding	**Evidence**
Moderate regime separation in latent space	PCA projections reveal partially separable low-dimensional behavioral regions; PC1 alone captures 80.3% of variance and isolates the *unstable* regime.
Behavioral features encode discriminative structure	All six features exhibit $F_k > 1$; `memory_usage` ($F = 4.66$) and `entropy` ($F = 4.13$) are most informative.
Linear discriminability is observable	LDA 5-fold CV accuracy of 70.4% ($\pm$0.2%) confirms jointly linearly separable behavioral regimes — 45 percentage points above chance.
Temporal behavioral structure is coherent	AR(1) emission dynamics produce smooth, autocorrelated trajectories with continuous regime transitions; empirical frequencies match the theoretical Markov stationary distribution within 0.5%.
Anomalous regimes are detectable	The *unstable* regime is flagged anomalous at 94.4% rate and exhibits a 5.1× higher error rate than the *stable* baseline.

3. **Anomaly detection (Notebook 04)** — Build an online Mahalanobis-distance anomaly scorer trained on the healthy-regime feature distributions; benchmark against the `is_anomaly` ground truth.
4. **Robustness analysis** — Vary the number of agents (N), timesteps (T), and random seeds to assess the stability of PCA variance capture and LDA accuracy under finite-sample regimes.

A. Computational Environment

Execution timestamps (UTC):

- Kernel start: `2026-05-07T17:28:42Z`
- Dataset generation: $\approx 1.7\,\mathrm{s}$
- Full notebook execution: $\approx 39\,\mathrm{s}$

Table 12: Software environment.

Component	Version
Python	3.12.13
NumPy	(standard scientific stack)
Pandas	(standard scientific stack)
Matplotlib	(standard scientific stack)
Seaborn	(standard scientific stack)
scikit-learn	(standard scientific stack)
SciPy	(standard scientific stack)
Notebook	Jupyter (IPython kernel)

B. File Manifest

Table 13: Files produced by Notebook 01.

File	Description
`data/telemetry_n20_t2000.csv`	Primary dataset
`data/X_telemetry.npy`	Feature matrix (numpy)
`data/y_labels.npy`	Label vector (numpy)
`fig01_state_freq.png`	State frequency bar chart
`fig02_feature_violins.png`	Feature distribution violins
`fig03_latency_entropy.png`	Primary behavioral plane scatter
`fig04_separability.png`	Fisher separability bar chart
`fig05_pca.png`	PCA 2-D projections
`fig06_scree.png`	PCA scree / cumulative variance
`fig07_temporal.png`	Temporal trajectory (agent_0000)
`fig08_centroid_dist.png`	Centroid distance heatmap